

Patan Multiple Campus
Pre-Board Examination-2083

Subject: Computer Graphics
Faculty: BIT
Semester: III/V

Time: 3:00 hrs.

F.M.: 60
P.M.: 24

SET-B (Sec-A)
Group – 'A'

Attempt any Two [2 x 10 = 20]

1. Write algorithm for Bresenham's Line Drawing algorithm. Generate the pixel in the line path with starting point (8, 10) and end point (12, 7) using Bresenham's line drawing algorithm.
2. Explain RGB color model. Differentiate between virtual reality and augmented reality with examples. Explain Binary Space Partitioning (BSP) trees with a suitable example.
3. Write the algorithm for Cohen-Sutherland Line clipping. Clip the polygon A(200,150), B(250,200) and C(400,300) with the clipping window defined by the coordinates (200,300), (400,300) and (500,100) using Sutherland Hodgeman Polygon Clipping algorithm.

Group-B

Attempt any EIGHT questions. [8 x 5 = 40]

5. What do you mean by the refresh rate of a display device? Explain vector scan display with a suitable diagram.
6. Plot the first octant of a circle centered at (-3, -3), having a radius of 5 units using the mid-point circle algorithm.
7. Given a triangle with vertices A(3,4), B(5,5), C(4,4) by rotating 90 degrees about the origin and then translating two units in each direction. Use the homogeneous transformation matrix to find the new vertices of the triangle.
8. Construct the Bezier curve with the following polygon vertices (control points): A(1,1), B(3,3), C(4,3), and D(8,4).
9. Compare the Scanline Method for Visible Surface Detection with the Depth Buffer Method.
10. What is Transformation. What are the role of Homogeneous coordinates?
11. Differentiate between parallel projection and perspective projections.
12. Explain polygon surface representation using Polygon table and polygon meshes.
13. Calculate the total memory required to store a 10 minute video in a SVGA system with 24 bit true color and 25 fps.
14. Write short notes on:
 - a) Open GL.
 - b) 3 D Object Representation

Tribhuvan University
Patan Multiple Campus
Bachelor in Information Technology (BIT)
Pre-Board Exam

Time: 3 hours (11:00 AM to 02:00 PM)

Subject: BIT302 Software Engineering

Semester: 5th

Total Marks: 60

Group – 'A'

Attempt any TWO Question.

[2 x 10 = 20]

1. What is System Modeling? What are different system models from different perspectives of the system? Explain. [3+7]
2. What is requirements in software engineering? Explain the Functional and Non-functional requirement of software. [3+7]
3. What do you understand by the term Software Testing? Explain the difference between Validation and Verification. [3+7]

Group – 'B'

Attempt any EIGHT Questions.

[8 x 5 = 40]

4. What views or perspectives are useful when designing and documenting a system's architecture? [5]
5. Explain the term Software engineering and System engineering? [5]
6. What is Black Box Testing? Explain the following Test Strategy used in Black box Testing. a)Equivalence Class Testing, b)Boundary Value Testing and c) Decision Table Testing [2+3]
7. What do you understand by the term "Test-driven development". Explain. [5]
8. Explain the differences between quality assurance and quality control. [5]
9. Software Project Management is different from other engineering project. Justify. [5]
10. Explain Process and Product quality standards of Software quality management. [5]
11. What is Configuration Management? Explain the Configuration management activities in brief. [1+4]
12. Write Short Notes. [2 x 2.5 = 5]
 - a. Reuse
 - b. Host-target development

-End-

Section A
Information Security (BIT303)
Group "A"

Attempt Any TWO questions:
[2×10=20]

13. What is Fiestel Structure? Explain DES round operation in detail.
14. What are the techniques used for method Authentication? Describe how 128-bit of hash value is generated by taking an input message of variable size using MD5 algorithm?
15. What is public key cryptography? Explain RSA algorithm with suitable numerical example.

Group "B"

Attempt Any EIGHT questions
[8×5=40]

16. What is IT Security Management? Explain any five security design principles in brief.
17. What is malicious code? Explain worms, virus, trojan horse and spyware in brief.
18. Divide $5x^2 + 4x + 6$ by $2x + 1$, assuming that the polynomials are over the field Z_7 i.e. $GF(7)$.
19. What is authentication and authorization? What is Biometric authentication? How it works?
20. Encrypt and decrypt the message "GENZ" using the Hill Cipher with the key $\begin{pmatrix} 3 & 3 \\ 2 & 5 \end{pmatrix}$.
Show your calculations and result.
21. Explain Diffie Hellman Key exchange protocol in brief.
22. Explain Role based access control in brief.
23. What is security Risk assessment? Explain different assessment approaches in brief.
24. Write short notes on:
 - c. Electronic transaction act 2063
 - d. Active and Passive attack

Patan Multiple Campus
Pre-Board Examination-2082

Subject: Web Technology-II
Faculty: BIT
Semester: III/V

Time: 3:00 hrs.

F.M.: 60
P.M.: 24

SET-B

Group – 'A'

Attempt any TWO Question.

[10 x 2 = 20]

- 1) How can you store numeric data in an array? Write a PHP program to demonstrate the use of array_push and array_pop functions for managing array elements. [3+7]
- 2) Differentiate between GET and POST method. How these two methods of data submission is handled in PHP? Explain with example (3+2+5)
3. a) How can you create a database connection? Create a PHP form with:
 - Text field
 - Password field
 - Radio button elementsb) Write a function to insert the form data into a database table. [2 + 8]

Group – 'B'

Attempt any EIGHT Questions.

[8 x 5 = 40]

- 4) How do you embed PHP code in web pages? Write a PHP program to illustrate the concepts of continue and break statements. (1+4)
- 5) Write a PHP program that will contain a concat_strings function. The function should have two arguments, st1 and st2 of type string, and should return st3, which is the concatenation of st1 and st2. [5]
- 6) How do you handle file upload errors in PHP? Write a PHP program that checks for upload errors and handles them accordingly. [5]
- 7) Define array and string. Create associative array of size seven and display the contents in descending order of key. (1+4)
- 8) Why file handling concept is used in PHP? Explain fwrite(), unlink() and fread() function with examples. (1+4)
- 9) What do you mean by Exception in PHP? Explain any two PHP built-in functions for Exception handling with example. [1+4]
- 10) Define Anonymous class in PHP. How can we create simple *index.php* form in PHP. [1+4]
- 11) What is HTTP authentication in PHP? WAP to simple HTTP authentication in PHP. [1+4]
- 12) What is JSON file in PHP? For the given BIT.CSV file:

Name, Age, Gender, Grade

Alice, 14, Female, 9

Bob, 15, Male, 10

Charlie, 13, Male, 8

Diana, 14, Female, 9

Ethan, 15, Male, 10

Write a PHP program to read and display this information.

Tribhuvan University
Patan Multiple Campus
Bachelor in Information Technology (BIT)
Pre-Board Examination- 2082

Subject: Technical Writing (ENG 305)
Semester: V
Time: 3 Hours

Total Marks: 60
Pass Marks: 24

Section A

Group – 'A' Long Answer Questions

Attempt any two Questions. $10 \times 2 = 20$

3. Write a letter to your campus chief requesting him to provide a hi-tech facility in your college library.
4. What is plagiarism? If plagiarism went unchecked, what would the consequences be for the student who plagiarizes, for students who don't, for the institution, for future employers, for the originator of the plagiarized material, and for society as a whole?

Group – 'B' Short Answer Questions

Attempt any eight questions. $8 \times 5 = 40$

9. Imagine that your software company has been in the market for two years. Write a short report on it.
10. What are the types of proposals? Briefly illustrate the parts of a proposal from front to back.
11. What is email etiquette? How does a technical writer apply proper netiquette to prevent email faux pas?
12. Define graphics with their various types? Describe its importance in technical writing.
13. An IT-based company is seeking to recruit an IT specialist. Compose an application letter tailored to that specific job ad so that you can impress your prospective employer.
14. Write an email to your foreign friend describing a product and its procedures to use it.
15. Suppose you are going to present a matter of issue in front of the audience. How do you make planning for your effective presentation? Explain.
16. Edit the sentences in the following exercises to make them as clear and concise as possible, without changing their meaning.
 - vi. There is a limitation to the effectiveness of on-site computer backups in the home or in the office because thieves may steal the back-up device at the very same time they steal a person's computer.
 - vii. It is an oft-told and generally accepted axiom that a 10,000-square-metre data centre has the capacity to release as much carbon dioxide as to be the equivalent of 100,000 SUVs on the road getting 25 litres per 100 kilometers.
 - viii. Given the fact that more and more employees are using and sharing confidential information on mobile devices such as tablets and smartphones, there is a growing trend that proprietary information is increasingly compromised due to lost or stolen mobile devices rather than any actual hacking.
 - ix. Malware in the form of Trojan horse viruses for stealing corporate information are increasingly being distributed through free games for children because it is well known that when employees bring home their work laptops, their children will oftentimes download games onto them to play.
 - x. What happened also is that because the streams and the rivers were once again shaded by trees, their temperature dropped and this had the resultant effect of allowing trout to return and indeed to thrive.

The End